01/05/2026

Code-A

Aakash
Medical | IIT-JEE | Foundations

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Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

**MM : 720** Pre-NEET for 2026 **Time : 180 Min.**

## PHYSICS

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## CHEMISTRY

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## BIOLOGY

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## Hints and Solutions

### PHYSICS

**(1)** **Answer :** (3)
**Solution:**
Potential due to dipole at a distance *r* is given by,

$V = \frac{k\vec{p}\cdot\vec{r}}{r^3}$

$= \frac{kp\cos\theta}{r^2}$

$= \frac{9\times10^9\times25\times10^{-9}\times\frac{3}{5}}{25\times10^{-2}}$

$= \frac{27}{5\times10^{-2}}$

= 540 V

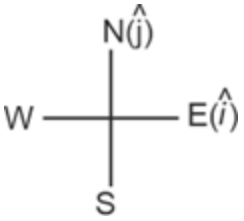
**(2)** **Answer :** (2)
**Solution:**

$B_0 = \frac{E_0}{c} = \frac{90}{3\times10^8} = 3\times10^{-7}\text{T}$

& $\vec{v} \left|\right| \vec{E}\times\vec{B}$

∴Magnetic field is along $\left(-\hat{k}\right)$

**(3)** **Answer :** (3)
**Solution:**




Force is parallel to $\Delta\vec{P}$

$P_i$ is along $\left(\hat{j}\right)$ and $P_f$ is along $\left(\hat{i}\right)$

$\Delta\vec{P} = \vec{P}_f - \vec{P}_i$

$\therefore \Delta\vec{P}$ is along $\left(\hat{i}-\hat{j}\right)$

i.e. along south-east

**(4)** **Answer :** (2)
**Solution:**

Average velocity $\left(\overline{V}_{\text{avg}}\right) = \frac{\text{Total displacement}}{\text{total time}}$

$= \frac{R}{T}$

$\Rightarrow R = \frac{2u_x u_y}{g}$ and $T = \frac{2u_y}{g}$

$\Rightarrow \overline{V}_{\text{avg}} = \frac{\frac{2u_x u_y}{g}}{\frac{2u_y}{g}}$

$= u_x$

$= 240\times\frac{4}{5}$

$= 192\ \text{m s}^{-1}$

**(5)** **Answer :** (1)
**Solution:**

$v_{\text{rms}} = \sqrt{\frac{3\,\text{RT}}{M}}$

$v_{\text{rms}} \propto \sqrt{T}$

$v_{\text{rms}}$ is increased by 2 times

i.e., final rms speed = $v + 2v = 3v$

$\Rightarrow \frac{V_2}{V_1} = \sqrt{\frac{T_2}{T_1}}$

$\frac{3v}{v} = \sqrt{\frac{T_2}{250}}$

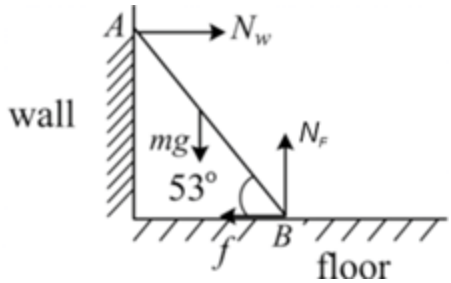
$\Rightarrow 9 = \frac{T_2}{250}$

$\Rightarrow T_2 = 2250\ \text{K}$

$= 1977^\circ\text{C}$

**(6) Answer :** (3)

**Solution:**




Balance force in vertical & horizontal direction,

$N_F = \text{mg}$ and $N_w = f$

Balancing torque about point *B*

$\text{mg}\,\frac{l}{2}\cos 53^\circ = N_w l \sin 53^\circ$

$\frac{mg}{2} \times \frac{3}{5} = N_w \times \frac{4}{5}$

$\Rightarrow N_w = \frac{3}{8}\text{mg}$— (ii)

$\Rightarrow f = N_w = \frac{3}{8} \times 50 \times 10 = 187.5\ \text{N}$

**(7) Answer :** (1)

**Solution:**

In a medium, speed of light $v = \frac{c}{\mu}$

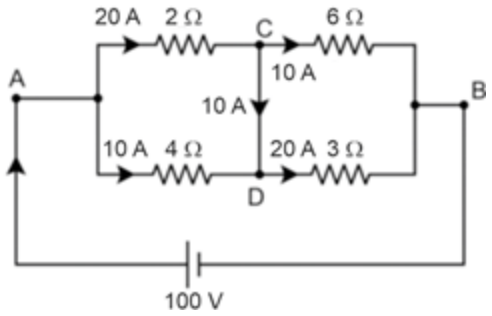
$\mu = a + \frac{b}{\lambda^2} \ldots$

$\lambda_{\text{orange}} > \lambda_{\text{blue}}$

$\therefore v_{\text{orange}} > v_{\text{blue}}$

**(8) Answer :** (1)

**Solution:**




$R_{AB} =$ $(2\ \Omega$ and $4\ \Omega)$ in series with $(6\ \Omega$ and $2\ \Omega)$

$= \frac{2\,\text{x}\,4}{2+4} + \frac{6\,\text{x}\,3}{6+3} = \frac{4}{3} + 2 = \frac{10}{3}\Omega$

$i = \frac{100}{R_{AB}} = \frac{100\,\text{x}\,3}{10} = 30\text{A}$

$\text{i}_{2\Omega} = \frac{4}{6} \times \text{i} = \frac{2}{3} \times 30 = 20$ A

$i_{4\Omega} = \frac{2}{6} i = \frac{1}{3} \times 30 = 10$ A

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$i_{6\Omega} = \frac{3}{9} \times i = \frac{1}{3} \times 30 = 10$ A

$i_{3\Omega} = 20$ A

**(9)** **Answer :** (1)

**Solution:**

$\rightarrow t = 0$ to $t = t$, acceleration is positive and constant $(a_0)$ $\therefore$ velocity increases

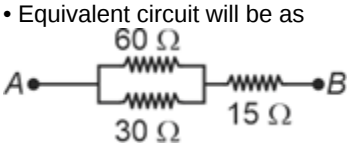
$\rightarrow t = t$ to $t = 2t$, acceleration is negative and constant and equal to $(-a_0)$ $\therefore$ velocity reduced to zero

$\rightarrow t = 2t$ to $3t$, acceleration is zero $\therefore$ velocity remains zero.

**(10)** **Answer :** (1)

**Solution:**

• Equivalent circuit will be as




hence $C_{eq}$ = 35Ω

• $E = \frac{12400}{4000} \approx 3$ eV

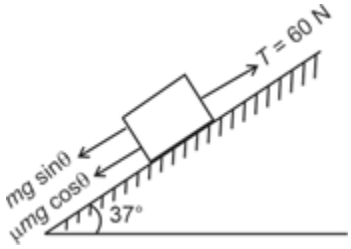
**(11)** **Answer :** (4)

**Solution:**

For a paramagnetic material, relative permittivity is slightly greater than 1.

**(12)** **Answer :** (3)

**Solution:**




$\Rightarrow T - (mg \sin\theta + \mu mg \cos\theta) = ma$

$\Rightarrow 60 - \left[4 \times 10 \times \frac{3}{5} + \frac{2}{10} \times 4 \times 10 \times \frac{4}{5}\right] = 4a$

$\Rightarrow 60 - [24 + 6.4] = 4a$

$\Rightarrow 29.6 = 4a$

$\Rightarrow a = 7.4 \text{ m s}^{-2}$

**(13)** **Answer :** (2)

**Solution:**

In $\gamma$-decay, both mass number and atomic number remain same

In $\beta^+$-decay, mass number remains same.

**(14)** **Answer :** (2)

**Solution:**

$E = \frac{kq}{r^2}$

$= \frac{9 \times 10^9 \times 8 \times 10^{-7}}{36 \times 10^{-4}}$

$= 2 \times 10^6$ N/C

**(15)** **Answer :** (4)

**Solution:**

$\left(\frac{dm}{dt}\right)_{ice} = 16\left(\frac{dm}{dt}\right)_{water}$

$\Rightarrow \frac{(\theta - 0)3kA}{80 \times 1} = \frac{16(\theta - 100)3kA}{560 \times 2l}$

$\Rightarrow \frac{\theta}{80} = \frac{16(\theta - 100)}{1120}$

On solving $\theta = 800^\circ$C

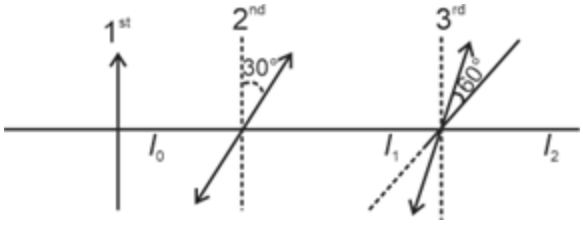
**(16)** **Answer :** (1)

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**Solution:**
From Bohr's model for hydrogen like atom we know that
$r \propto \frac{n^2}{z},\ v \propto \frac{z}{n},\ L \propto n,\ E \propto \frac{z^2}{n^2}$

**(17) Answer :** (1)
**Solution:**
$Y = \overline{\overline{A \cdot B} + \overline{A \cdot B}}$
$= \overline{\overline{A \cdot B}} \cdot \overline{\overline{A \cdot B}}$
$= (A \cdot B) \cdot (A \cdot B)$
$= A \cdot B$
= AND gate

**(18) Answer :** (2)
**Solution:**




$I_1 = I_0 \cos^2 30^\circ = I_0 \times \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3I_0}{4}$
$I_2 = I_1 \cos^2 60^\circ = \frac{3I_0}{4} \times \left(\frac{1}{2}\right)^2$
$= \frac{3I_0}{16}$

**(19) Answer :** (1)
**Solution:**
Magnetic field induction at the centre of current carrying loop $B = \frac{\mu_0 i}{2R}$
$B \propto \frac{1}{R}$
$\frac{B_1}{B_2} = \frac{R_2}{R_1}$
$= \frac{3}{1}$

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**(20) Answer :** (3)
**Solution:**
$P_1V_1 + P_2V_2 = P(V_1 + V_2)$
$P = \frac{P_1V_1 + P_2V_2}{V_1 + V_2}$
$= \frac{1.4 \times 10 + 0.8 \times 15}{25}$
$= \frac{14 + 12}{25}$
$= \frac{26}{25}$
$= 1.04\ \text{atm}$

**(21) Answer :** (2)
**Solution:**
$Z = \sqrt{R^2 + (X_L - X_C)^2}$
$Z = \sqrt{(60)^2 + (80 - 20)^2}$
$Z = 60\sqrt{2}\ \Omega$
Power factor, $\cos\phi = \frac{R}{Z}$
$= \frac{60}{60\sqrt{2}}$
$= \frac{1}{\sqrt{2}}$
≈0.707

**(22) Answer :** (4)
**Solution:**

Maximum acceleration, $a_{\max} = w^2 A$

& $w = \sqrt{\frac{k}{m}}$

$a = \frac{kA}{m}$

$a_P = a_Q$

$k_P A_P = k_Q A_Q$

$\frac{A_Q}{A_P} = \frac{k_P}{k_Q}$

**(23) Answer :** (1)

**Solution:**

Initially, fundamental frequency of pipe open at both ends, $f = \frac{v}{2l}$

Now, $70\%$ length is dipped in water

$\therefore$ remaining length of air column, $l_c = \frac{30}{100} l = \frac{3l}{10}$

$\therefore$ fundamental frequency $l'$, $f' = \frac{v}{4l_c}$ (closed at open end)

$= \frac{v}{4\left(\frac{3l}{10}\right)} = \frac{5}{6} f$

**(24) Answer :** (2)

**Solution:**

Apply energy conservation

$\frac{1}{2} m v_0^2 = \frac{1}{2} m u^2 + mgh$

$\Rightarrow v_0^2 = u^2 + 2gh$ ...(i)

$\Rightarrow$ At point of slack, $mg\cos\alpha = \frac{mu^2}{l}$

$\Rightarrow u^2 = gl\cos\alpha$ — (ii)

By eq (i) and (ii)

$v_0^2 = gl\cos\alpha + 2g\left(l + l\cos\alpha\right)$

$v_0^2 = gl\cos\alpha + 2gl + 2gl\cos\alpha = 2gl + 3gl\cos\alpha$

$\frac{v_0^2}{gl} = \left(2 + 3\cos\alpha\right)$

**(25) Answer :** (4)

**Solution:**

Assuming the planet to be solid sphere, $I = \frac{2}{5}\mathrm{mR}^2$

using conservation of angular momentum,

$I'\omega' = \mathrm{I}\omega$

$\Rightarrow \frac{2}{5} m\left(3R\right)^2 \times \frac{2\pi}{T'} = \frac{2}{5}\mathrm{mR}^2 \times \frac{2\pi}{T}$

$\Rightarrow \frac{9}{T'} = \frac{1}{T}$

$\Rightarrow T' = 9T$

$= 9 \times 15$

$= 135$ days

**(26) Answer :** (2)

**Solution:**

Maximum percentage error in $P = \left(\frac{1}{2}\frac{\Delta a}{a} + 2\frac{\Delta b}{b} + 2\frac{\Delta c}{c} + \frac{2}{3}\frac{\Delta d}{d}\right) \times 100$

$= \frac{1}{2} \times 2 + 2 \times 1 + 2 \times \frac{1}{2} + \frac{2}{3} \times 3$

$= 1 + 2 + 1 + 2$

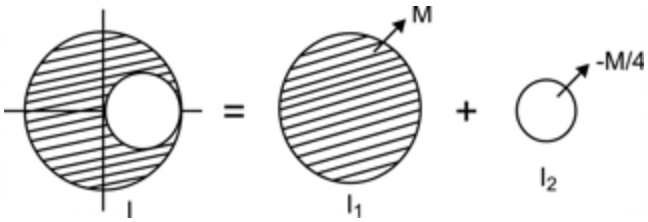
$= 6\%$

**(27) Answer :** (1)

**Solution:**

Mass of the disc that is cut = $\frac{M}{\pi(2R)^2} \times \pi R^2 = \frac{M}{4}$

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:

$I = I_1 + I_2$

$I = \frac{M(2R)^2}{2} + \left[\frac{\left(-\frac{M}{4}\right)R^2}{2} + \left(-\frac{M}{4}\right)R^2\right]$

$I = 2MR^2 - \left[\frac{3}{8}MR^2\right] = \frac{13}{8}MR^2$

**(28) Answer :** (1)
**Solution:**

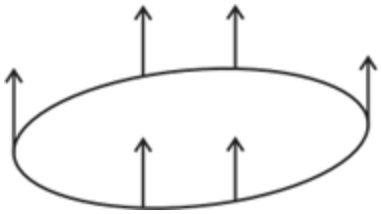
maximum strain = $\frac{\text{maximum stress}}{\text{Young's modulus}}$

$\frac{\Delta L}{L} = \frac{1.2\times10^8}{8\times10^{10}}$

$\Delta L = \frac{1.2}{8} \times 10^{-2} \times \frac{1}{2}$

$= 0.75 \text{ mm}$

**(29) Answer :** (2)
**Solution:**



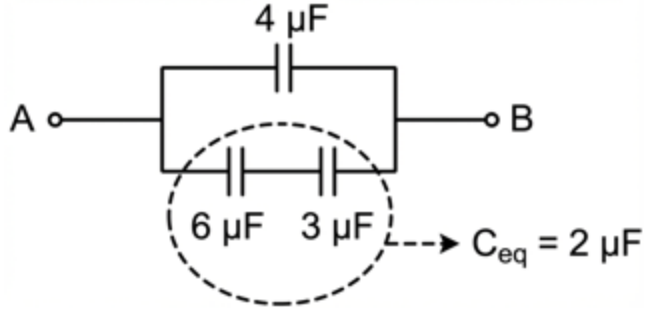
Excess force = $2T \times 2\pi R$

$= 2 \times \frac{7}{100} \times 2 \times \frac{22}{7} \times \frac{5}{100}$

$= 44 \text{ mN}$

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**(30) Answer :** (2)
**Solution:**
$2\ \mu\text{F}$ capacitor is short circuited.
∴circuit can be redraw as




$C_{AB} = 4 + 2 = 6\ \mu\text{F}$

**(31) Answer :** (4)
**Solution:**
Path *bc* is an isobaric process and volume increases
$\therefore W = P\,\Delta V$

$= 3 \times 10^5 \left[400 - 100\right] \times 10^{-6}$

$= 3 \times 30$

$= +90\,\text{J}$

**(32) Answer :** (3)
**Solution:**
$\Delta E = \frac{hc}{\lambda}$
$\Rightarrow \frac{hc}{\lambda_1} = 7E$ ...(i)
$\Rightarrow \frac{hc}{\lambda_2} = 4E$ ...(ii)
$\Rightarrow \frac{hc}{\lambda_3} = 5E$ ...(iii)
on comparing eq (i) and (ii)
$\frac{\lambda_2}{\lambda_1} = \frac{7}{4} \Rightarrow \lambda_2 = \frac{7}{4}\lambda_1$
on comparing eq (ii) and (iii)
$\frac{\lambda_3}{\lambda_2} = \frac{4}{5} \Rightarrow \lambda_3 = \frac{4}{5}\lambda_2$
and $E_1 > E_3 > E_2$
$\Rightarrow \lambda_1 < \lambda_3 < \lambda_2$

**(33) Answer :** (4)
**Solution:**
Microwaves has largest wavelength among the following.

**(34) Answer :** (2)
**Solution:**
Least count = $\frac{\text{Pitch}}{\text{No. of circular divisions}}$
= $\frac{0.6}{50}$
= $0.012\ \text{mm}$
Reading = MSR + (L.C.) x No. of circular division
= $4 + 0.012 \times 40$
= $4.480\ \text{mm}$

**(35) Answer :** (3)
**Solution:**
Inductance $L = \mu_0 n^2 V$
Inductance depends on geometry and medium inside the inductor.

**(36) Answer :** (2)
**Solution:**
Moment of inertia about tangential axis for solid uniform sphere $= \frac{7}{5}mR^2$
$\Rightarrow \frac{7}{5} \times 12 \times R^2 = 336$
$\Rightarrow R^2 = \frac{336 \times 5}{12 \times 7} = 20$
$\Rightarrow R = \sqrt{20}\ \text{m}$
$\Rightarrow \frac{6x}{\sqrt{5}} = 2\sqrt{5}$
$\Rightarrow x = \frac{2 \times 5}{6} = \frac{10}{6} = 1.67$

**(37) Answer :** (2)
**Solution:**
$v = \sqrt{2gh}$
$\Rightarrow \frac{v_2}{v_1} = \sqrt{\frac{0.36h}{h}} = 0.6$
$\Rightarrow v_2 = 0.6v_1$
$\Rightarrow \frac{(K.E.)_2}{(K.E.)_1} = \frac{{v_2}^2}{{v_1}^2} = 0.36$
$\Rightarrow (K.E.)_2 = 0.36(K.E.)_1$
$\Rightarrow \%\ \text{lost} = 64\%$

**(38) Answer :** (1)
**Solution:**
For transformer,
$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{220}{11000} = \frac{1}{50}$
$\Rightarrow N_s = \frac{N_p}{50} = \frac{5000}{50} = 100$
For ideal transformer,

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$P_{\text{in}} = P_{\text{out}}$
$i_s V_s = i_p V_p$
$\Rightarrow i_p = \frac{25\times220}{11000} = 0.5\text{A}$

**(39) Answer :** (3)
**Solution:**
W.D. = $\Delta K.E.$
$\Rightarrow \vec{F} \cdot \vec{S} = k_f - 25$
$\Rightarrow \left(-4\hat{i}+3\hat{j}\right)\cdot\left(5\hat{i}+2\hat{j}\right) = k_f - 25$
= $-20+6 = k_f - 25$
$\Rightarrow k_f = 11\text{ J}$

**(40) Answer :** (2)
**Solution:**
de-Broglie wavelength is given by
$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqV}}$
For same potential difference
$\lambda \propto \frac{1}{\sqrt{\text{mq}}}$
$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}} = \sqrt{\frac{4m_p\times 2e}{m_p e}}$
$\frac{\lambda_p}{\lambda_\alpha} = 2\sqrt{2}$
$\lambda_p = 2\sqrt{2}\,\lambda_\alpha$

**(41) Answer :** (2)
**Solution:**
$v_e = \sqrt{\frac{2GM}{R}}$
$v_e = \sqrt{\frac{2G\times\frac{4}{3}\pi R^3\rho}{R}}$
$v_e \propto R\sqrt{\rho}$
$v_e' = 12\,v$

**(42) Answer :** (3)
**Solution:**
$R_{net} = \int_0^L \frac{a_0 x\,dx}{A} = \frac{a_0}{A}\left[\frac{x^2}{2}\right]_0^L = \frac{a_0}{A}\frac{L^2}{2}$
$I = \frac{V}{R_{net}} = \frac{20(2A)}{a_0L^2} = \frac{40A}{a_0L^2}$
$\Rightarrow R_x = \int_0^x \frac{a_0 x\,\text{dx}}{A} = \frac{a_0x^2}{2A}$
$V_x = IR_x$
$V_x = \frac{40A}{a_0L^2}\times\frac{a_0x^2}{2A} = \frac{20x^2}{L^2}$
$\left|\vec{E}\right| = \frac{dV}{dx} = \frac{20}{L^2}2x$
$\left|\vec{E}\right|_{at\ x=20cm} = \frac{20}{1^2}\times\frac{2\times20}{100}$
=8 V/m

**(43) Answer :** (3)
**Solution:**
Flux through loop = $\pi R^2(a+bt)$
$\phi = \pi R^2 a + \pi R^2 bt$
$\left|E_{\text{ind}}\right| = \frac{d\phi}{dt} = \pi R^2 b$

**(44) Answer :** (2)
**Solution:**

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Power = $F_v \times v_t$
= $6\pi\eta r v_t^2$
& $v_t \propto r^2$
$\therefore P \propto r^5$
$\therefore$ 32 times.

**(45) Answer :** (3)
**Solution:**
inside ( $d < R$), $B = \frac{\mu_0 i}{2\pi R^2} d$ i.e. $B \propto d$
outside ( $d > R$)
$B = \frac{\mu_0 i}{2\pi d}$
i.e. $B \propto \frac{1}{d}$

**CHEMISTRY**

**(46) Answer :** (2)
**Solution:**
$[\mathrm{Mn(CN)_6}]^{3-}$ and $[\mathrm{Fe(CN)_6}]^{3-}$ are inner orbital complexes involving $d^2sp^3$ hybridisation. These are paramagnetic in nature.

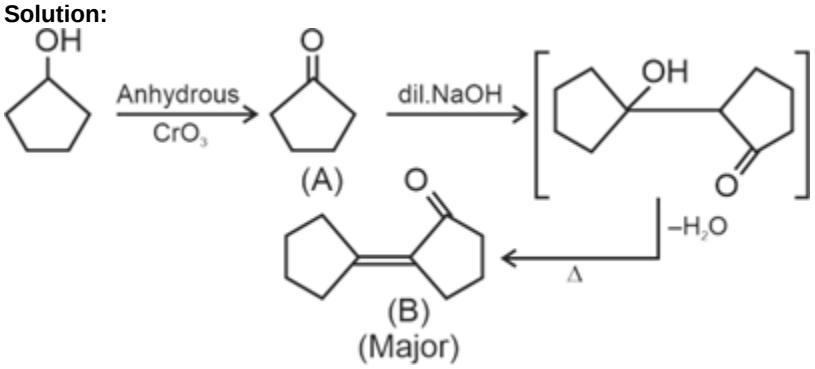
**(47) Answer :** (3)
**Solution:**
$8\,\mathrm{MnO_4^-} + 3\mathrm{S_2O_3^{2-}} + \mathrm{H_2O} \rightarrow 8\,\mathrm{MnO_2} + 6\,\mathrm{SO_4^{2-}} + 2\,\mathrm{OH^-}$
2 moles of $\mathrm{S_2O_3^{2-}} = \frac{8\times2}{3}$ moles of $\mathrm{MnO_4^-}$

**(48) Answer :** (4)
**Solution:**

| **Ion** | Number of unpaired electron(s) | **Spin only magnetic moment value in BM** |
|---|---|---|
| $\mathrm{Ni^{2+}}$ | 2 | 2.83 |
| $\mathrm{Cr^{2+}}$ | 4 | 4.90 |
| $\mathrm{Ti^{3+}}$ | 1 | 1.73 |
| $\mathrm{Co^{2+}}$ | 3 | 3.87 |

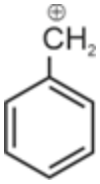
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**(49) Answer :** (1)
**Solution:**
$(\mathrm{NH_4})_2\,\mathrm{Cr_2\,O_7} \xrightarrow{\Delta} \mathrm{N_2} + 4\mathrm{H_2O} + \mathrm{Cr_2\,O_3}$

**(50) Answer :** (2)
**Solution:**



**(51) Answer :** (3)
**Solution:**
188 g of AgBr contains 80g bromine
0.235 g of AgBr contains $\frac{80\times0.235}{188}$ g bromine
Percentage of bromine = $\frac{80\times0.235\times100}{188\times0.25} = 40\%$

**(52)** **Answer :** (3)
**Solution:**

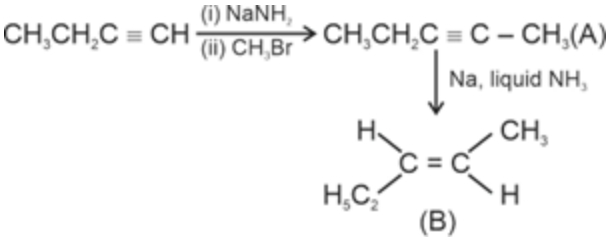


does not show hyperconjugation effect

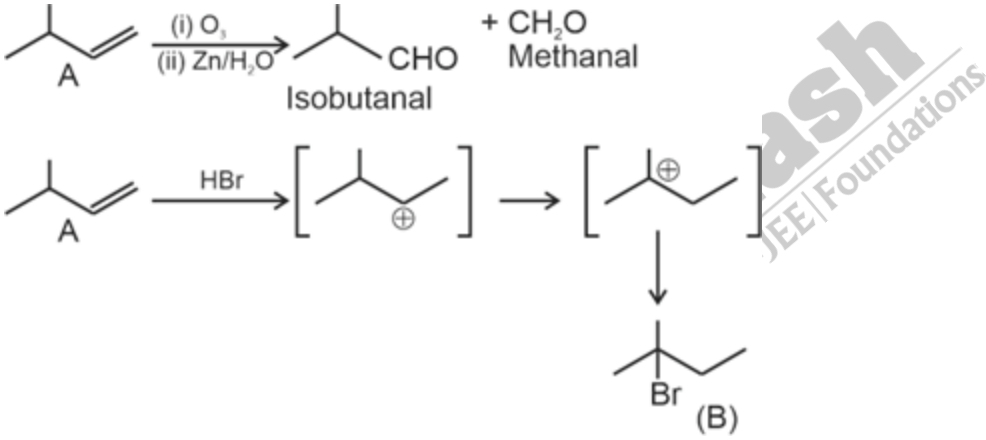
**(53)** **Answer :** (3)
**Solution:**

- $[Fe(SCN)]^{2+}$ is blood red in colour
- $NaSCN + 2Na \rightarrow NaCN + Na_2S$

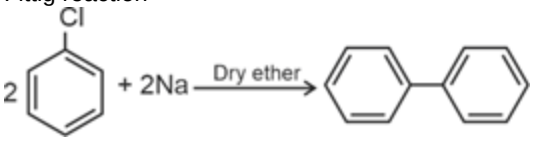
**(54)** **Answer :** (2)
**Solution:**



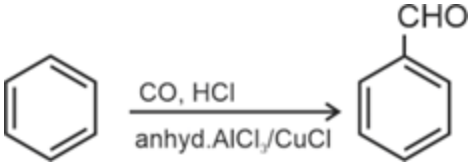

**(55)** **Answer :** (3)
**Solution:**



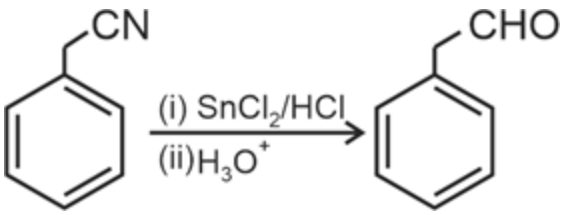

**(56)** **Answer :** (1)
**Solution:**
Fittig reaction



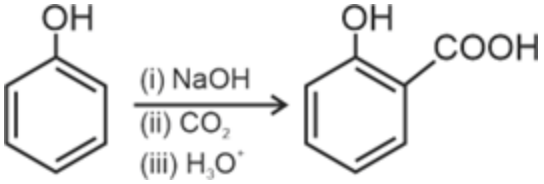

Gatterman-Koch reaction




Stephen reaction



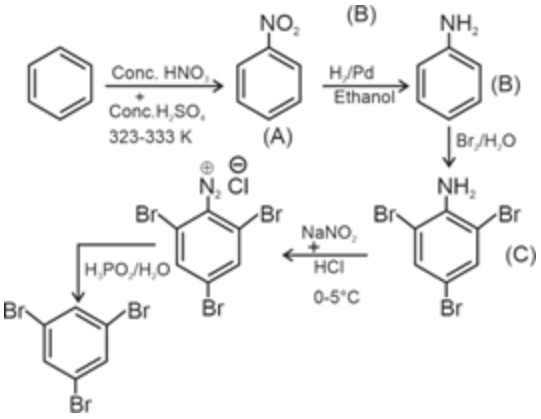

Kolbe's reaction




**(57)** **Answer :** (1)
**Solution:**
Correct order of acidic strength of the acids: $\mathrm{CF_3\,COOH > NO_2\,CH_2\,COOH > ClCH_2\,COOH > PhCOOH}$

**(58)** **Answer :** (1)
**Solution:**
In aniline, lone pair of nitrogen is delocalised in the benzene ring, hence it is less available for protonation.

**(59)** **Answer :** (3)
**Solution:**

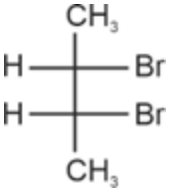



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**(60)** **Answer :** (2)
**Solution:**
$\mathrm{SiO_2 + 2NaOH \rightarrow Na_2\,SiO_3 + H_2O}$

**(61)** **Answer :** (2)
**Solution:**
Electron withdrawing group present at ortho and para position will stabilize the intermediate carbanion which is formed by the attack of $OH^-$ ion and will accelerate the substitution reaction.

**(62)** **Answer :** (2)
**Solution:**




Contains plane of symmetry hence it is optically inactive

**(63)** **Answer :** (4)
**Solution:**

| | |
|---|---|
| $XeF_4$ | Square planar |
| $CCl_4$ | Tetrahedral |
| | Trigonal bipyramidal |

| | |
|---|---|
| $PCl_5$ | |
| $BrF_5$ | Square pyramidal |

**(64)** **Answer :** (2)
**Solution:**
$N_2^+$ (electronic configuration):
$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2\left(\pi 2p_x^2=\pi 2p_y^2\right)\ (\sigma 2p_z)^1$

**(65)** **Answer :** (2)
**Solution:**

| Element | Electronegativity (On Pauling scale) |
|---|---|
| C | 2.5 |
| P | 2.1 |
| Si | 1.8 |
| Al | 1.5 |

**(66)** **Answer :** (2)
**Solution:**
Power of bulb = 30 watt = $30\ Js^{-1}$
Energy of photon = $\frac{hc}{\lambda}=\frac{6.6\times10^{-34}\times3\times10^8}{1320\times10^{-9}}=1.5\times10^{-19}\ J$
Number of photons emitted per second = $\frac{30}{1.5\times10^{-19}}=20\times10^{19}=2\times10^{20}$

**(67)** **Answer :** (3)
**Solution:**
$r\propto\frac{n^2}{Z}$
$\mathrm{r}_2\left(\mathrm{Li}^{2+}\right)\propto\frac{(2)^2}{3}$ ...(i)
$\mathrm{r}_3\left(\mathrm{He}^{+}\right)\propto\frac{(3)^2}{2}$ ...(ii)
$\frac{\mathrm{r}_3\left(\mathrm{He}^{+}\right)}{\mathrm{r}_2\left(\mathrm{Li}^{2+}\right)}=\frac{9}{2}\times\frac{3}{4}=\frac{27}{8}$
$\mathrm{r}_3\left(\mathrm{He}^{+}\right)=\frac{27}{8}\mathrm{r}_2\left(\mathrm{Li}^{2+}\right)$
$=\frac{27}{8}\mathrm{x}\ \mathrm{pm}$

**(68)** **Answer :** (3)
**Solution:**
$X\left(\mathrm{av}\right)=\frac{54\times70+53\times20+55\times10}{100}$
$=\frac{3780+1060+550}{100}=53.9\mathrm{u}$

**(69)** **Answer :** (2)
**Solution:**
$\Delta H=\Delta U+\Delta n_g\,\mathrm{RT}$
$\Delta H=4.2+3\times2\times300\times10^{-3}$
$=4.2+1.8\ =6.0$
$\Delta G=\Delta H-T\,\Delta\,S$
$=6-300\times40\times10^{-3}$
$=6-12$
$\Delta G=-6\ \mathrm{kcal}$

**(70)** **Answer :** (2)
**Solution:**
$CaCl_2$ is a strong electrolyte
$\mathrm{AgCl}\left(\mathrm{s}\right)\rightleftharpoons\underset{\mathrm{s}}{\mathrm{Ag}^{+}(\mathrm{aq})}+\underset{(\mathrm{s}+0.04)\simeq0.04}{\mathrm{Cl}^{-}(\mathrm{aq})}$

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$[Ag^+][Cl^-] = K_{sp}(AgCl)$

$s \times 0.04 = 1.8 \times 10^{-10}$

$s = \frac{1.8 \times 10^{-10}}{0.04} = 45 \times 10^{-10} = 4.5 \times 10^{-9}$ M

**(71) Answer :** (4)
**Solution:**

| Vitamin | Deficiency disease |
|---|---|
| Ascorbic acid | Scurvy |
| Vitamin E | Increased fragility of RBCs |
| Thiamine | Beri beri |
| Pyridoxine | Convulsions |

**(72) Answer :** (3)
**Solution:**
Amylopectin is insoluble in water.
Number of chiral carbons in Sucrose is 9.

**(73) Answer :** (2)
**Solution:**
In case of positive deviation from Raoult's law, A-B interactions are weaker than those between A-A or B-B interactions.

**(74) Answer :** (4)
**Solution:**

$$\begin{array}{lccccc} & \mathrm{HA} & \rightleftharpoons & \mathrm{H^+} & + & \mathrm{A^-} \\ \mathrm{At\ \ equilibrium} & 1-\alpha & & \alpha & & \alpha \end{array}$$

$i = 1 + \alpha = 1.6$
$\Delta T_f = iK_f m$
$= 1.6 \times 1.86 \times 0.2$
$\Delta T_f = 0.6$
Freezing point of solution = –0.6°C

**(75) Answer :** (1)
**Solution:**
Mercury cell consists of zinc-mercury amalgam as anode and a paste of HgO and carbon as cathode. The electrolyte used is a paste of KOH and ZnO.

**(76) Answer :** (2)
**Solution:**

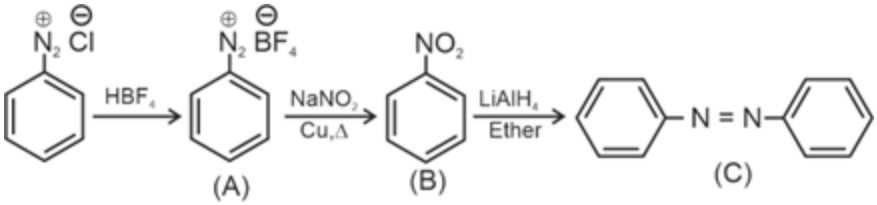
| Ion | $\lambda°$(S cm$^2$ mol$^{-1}$) in water 298 K |
|---|---|
| $Ca^{2+}$ | 119 |
| $Mg^{2+}$ | 106 |
| $Cl^-$ | 76.3 |
| $Br^-$ | 78.1 |

**(77) Answer :** (1)
**Solution:**

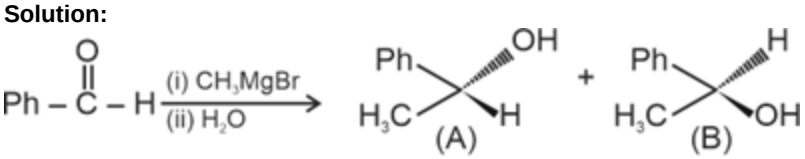
$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$
$= \frac{2.303}{50 \times 60} \log \frac{0.2}{0.0125}$
$= \frac{2.303}{3000} \log 16$
$k = \frac{2.303}{3000} \times 4 \times 0.3 = 9.2 \times 10^{-4} s^{-1}$

**(78) Answer :** (3)
**Solution:**

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**(79) Answer :** (2)
**Solution:**




(A) and (B) are a pair of enantiomers. They are non-superimposable mirror images.

**(80) Answer :** (2)
**Solution:**
$Al^{3+}$ and $Fe^{3+}$ belong to group-III in qualitative analysis of cations.

**(81) Answer :** (2)
**Solution:**

| Oxide | Chemical nature |
|---|---|
| $GeO_2$ | Acidic |
| PbO | Amphoteric |
| $N_2O$ | Neutral |
| CaO | Basic |

**(82) Answer :** (1)
**Solution:**

| Element | $\Delta_{eg}H$ (kJ mol$^{-1}$) |
|---|---|
| O | –141 |
| S | –200 |
| Li | –60 |
| Na | –53 |

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**(83) Answer :** (4)
**Solution:**
$4Zn + 10\,HNO_3(dilute) \rightarrow 4\,Zn(NO_3)_2 + 5H_2O + N_2O$
$3Cu + 8\,HNO_3(dilute) \rightarrow 3\,Cu(NO_3)_2 + 4H_2O + 2\,NO$

**(84) Answer :** (1)
**Solution:**

| | $CH_3COOH(aq)$ + | $NaOH(aq)$ ⟶ | $CH_3COONa$ + | $H_2O$ |
|---|---|---|---|---|
| Initial | 100×0.08 = 8 mmol | 400×0.02 = 8 mmol | – | – |
| Final | 0 | 0 | 8 mmol | |

$\left[CH_3COONa\right] = \frac{8}{500} = 0.016\ M$
Salt of weak acid and strong base will undergo hydrolysis
$pH = 7 + \frac{1}{2}\left[pK_a + \log\ c\right]$
$= 7 + \frac{1}{2}\left[4.76 + \log\ 0.016\right]$
$= 7 + \frac{1}{2}\left[4.76 + \log\ 16 \times 10^{-3}\right]$
$pH = 7 + \frac{1}{2}\left[4.76 - 3 + 4\log\ 2\right]$
$= 7 + 1.48$
$pH = 8.48 \simeq 8.5$

**(85)** **Answer :** (2)
**Solution:**
If inert gas (helium) is introduced to a reaction at equilibrium at constant pressure then the reaction will move in the direction where number of moles of the gaseous molecules increases.

**(86)** **Answer :** (4)
**Solution:**

| Ion | Ionic radii (pm) |
|---|---|
| $Nd^{3+}$ | 99 |
| $Eu^{3+}$ | 95 |
| $Dy^{3+}$ | 91 |
| $Yb^{3+}$ | 86 |

**(87)** **Answer :** (3)
**Solution:**
In a disproportionnation reaction an element in one oxidation state in simultaneously oxidised and reduced

**(88)** **Answer :** (4)
**Solution:**

| List I (Metal) | List II [Most common oxidation state(s)] |
|---|---|
| Ti | +4 |
| V | +5 |
| Cr | +3 and +6 |
| Ni | +2 |

**(89)** **Answer :** (1)
**Solution:**
Standard molar enthalpy of formation of $Br_2(l)$ is zero at 298 K

**(90)** **Answer :** (4)
**Solution:**
The thermal decomposition of HI on gold surface is an example of zero order reaction.

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## BIOLOGY

**(91)** **Answer :** (2)
**Solution:**
Due to insertion of β-galactosidase gene at the restriction site of *Eco*RI , the recombinant colonies will be blue in colour.
Because alien piece of DNA is inserted at *Bam* HI site in $tet^R$ gene, it causes insertional inactivation of $tet^R$ gene. So, recombinant colonies will be sensitive to tetracycline but still will be resistant to ampicillin. So, recombinant colonies having an alien piece of DNA will grow on medium containing ampicillin and will give blue colour in presence of a chromogenic substrate *i.e.*, X-gal.

**(92)** **Answer :** (1)
**Solution:**
The protein part of a holoenzyme is called apoenzyme and non-protein part is called co-factor which is important for biochemical activity of enzyme for a particular reaction. Co-factors may be organic (prosthetic group, co-enzyme) or inorganic (metal ions). When co-factor (organic compounds such as prosthetic group or inorganic compounds such as metal ions) are tightly bound to apoenzyme, it is called prosthetic group. Association of co-enzymes with apoenzyme is only transient, usually occurring during the course of catalysis. Most of the co-enzymes are vitamins, or vitamin derivatives.

**(93)** **Answer :** (1)
**Solution:**
Plants capture only 2-10% of the PAR and this small amount of energy sustains the entire living world.
Primary production is defined as the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis.

**(94)** **Answer :** (1)
**Solution:**

The synergids found at the micropylar end, have special cellular thickenings at the micropylar tip called filiform apparatus.

**(95) Answer :** (4)
**Solution:**
Neoplastic cells lose the property of contact inhibition. They are a mass of proliferating cells, invading and damaging the surrounding normal tissues. They compete with normal cells for vital nutrients. The cells sloughed from malignant tumors reach distant sites through blood and get lodged in the body at new sites to form a new tumor. This property of neoplastic cells is called metastasis.

**(96) Answer :** (2)
**Solution:**
Seeds are absent in pteridophytes. They have independent sporophyte and gametophyte.

**(97) Answer :** (2)
**Solution:**

| | |
|---|---|
| Estrogen | Ovary |
| Aldosterone | Adrenal cortex |
| Gonadotropin | Pars distalis |
| Somatostatin | Hypothalamus |

**(98) Answer :** (1)
**Solution:**
In human genome, chromosome Y has the lowest number of genes, *i.e.* ,231.

**(99) Answer :** (2)
**Solution:**
PEPcase catalyses the carboxylation of phosphoenol pyruvate (primary $CO_2$ acceptor) in $C_4$ plants. This enzyme is found in mesophyll cells of $C_4$ plant.

**(100) Answer :** (3)
**Solution:**
The first menstruation begins at puberty and is called menarche. The menstrual cycle starts with the menstrual phase, when menstrual flow occurs and it lasts for 3-5 days. In human beings, menstrual cycle ceases around 50 years of age; that is termed as menopause. Reproductive cycle in female primates is called menstrual cycle.

**(101) Answer :** (3)
**Solution:**
In 1983, Eli Lilly, an American company, prepared two DNA sequences corresponding to A and B-chains of human insulin and introduced them in plasmids of *E. coli* to produce insulin chains. Disarmed retroviruses are widely used as vectors in genetic engineering, especially for gene therapy in animal cells. *Agrobacterium tumefaciens* is considered as natural engineer of plants.

**(102) Answer :** (2)
**Solution:**
Sub-phylum Vertebrata is divided into two divisions-Agnatha and Gnathostomata, so, all gnathostomes are vertebrates but all vertebrates are not gnathostomes.

**(103) Answer :** (3)
**Solution:**
Centrosome helps in the assemblage of mitotic apparatus.

**(104) Answer :** (1)
**Solution:**
The unequivocal proof that DNA is the genetic material came from the experiment of Hershey and Chase. Avery, MacLeod and McCarty provided the biochemical characterisation of transforming principle. Taylor *et al.* used *Vicia faba* as the experimental organism. Meselson and Stahl used *E. coli* as the experimental organism.

**(105) Answer :** (3)
**Solution:**
Adenosine is a nucleoside containing adenine and ribose sugar moiety. Uracil is a pyrimidine derivative nitrogenous base. Glycine is the simplest α-amino acid containing an achiral carbon atom attached with two hydrogen atoms. Uridylic acid is a nucleotide formed by uracil, pentose sugar and phosphate moiety.

**(106) Answer :** (4)
**Solution:**
Special venous connection between liver and intestine as well as kidney and lower parts of the body are present in frogs. The former is called hepatic portal system and latter one is called renal portal system. Renal veins drain blood from kidneys to heart through vena cava.

**(107) Answer :** (4)

**Solution:**

RNA polymerase binds to promoter and initiates transcription. RNA polymerase polymerises ribonucleotides in template dependent manner. Termination of transcription in prokaryotes is facilitated by ρ factor. hnRNA is formed in eukaryotes.

**(108) Answer :** (2)

**Solution:**

Polymerase chain reaction (PCR) amplifies the DNA as per the equation $2^n$ where 'n' is the number of PCR cycles.

So here, $1024 = 2^n$

So, n = 2 × 2× 2 × 2× 2 × 2× 2 × 2× 2 × 2

So, $1024 = 2^{10}$

So, n = 10 cycles

**(109) Answer :** (4)

**Solution:**

Pollen grains coming in contact with the stigma is a chance factor in both wind and water pollination. To compensate for this uncertainties and associated loss of pollen grains, the flowers produce enormous amount of pollen when compared to the number of ovules available for pollination.

**(110) Answer :** (3)

**Solution:**

The cattle egret and grazing cattle in close association is an example of commensalism.

**(111) Answer :** (3)

**Solution:**

In human females, parturition is a complex neuroendocrine mechanism involving neural system along with cortisol, estrogen, and oxytocin. The signals for parturition originate from the fully developed foetus and placenta, which induce mild uterine contractions, called foetal rejection reflex. Cervical canal and vagina together constitute the birth canal. Uterus is not the part of birth canal.

**(112) Answer :** (4)

**Solution:**

'A' is the site through which acid/base is added for pH control. 'B' is motor which is used for rotation of foam breaker and impeller. 'C' is culture broth used for culturing microorganisms for synthesis of proteins. 'D' is used to provide sterile air for aerobic conditions.

**(113) Answer :** (4)

**Solution:**

Frogs undergo aestivation during peak summer and hibernation during peak winter. They perform cutaneous respiration during summer sleep (aestivation) as well as during winter sleep (hibernation).

**(114) Answer :** (3)

**Solution:**

$T_4$ and $T_3$ are secreted by thyroid follicles. They regulate rhythm of menstrual cycle, support the process of erythropoiesis and maintain of water and electrolyte balance of the body. Calcium metabolism is regulated by thyrocalcitonin which is synthesised by para follicular cells of thyroid gland. Hypersecretion of parathyroid hormone causes osteoporosis.

**(115) Answer :** (3)

**Solution:**

ABA stimulates the closure of stomata. Apical dominance is promoted by IAA.

**(116) Answer :** (1)

**Solution:**

Somatostatin is an inhibitory neuropeptide secreted from hypothalamus and inhibits secretion of growth hormone. Vasopressin and oxytocin are also neuropeptides but reach to posterior pituitary directly through axons of neurosecretory cells. Gonadotropins do not enter in portal circulation.

**(117) Answer :** (1)

**Solution:**

Acetic acid is derived from bacteria-*Acetobacter aceti.*

**(118) Answer :** (4)

**Solution:**

Gene controlling flower colour in Snapdragon shows incomplete dominance (Non-Mendelian inheritance pattern).

**(119) Answer :** (3)

**Solution:**

| | |
|---|---|
| Acrosome | Contains secretion that helps the sperm enter into the ovum |

| | |
|---|---|
| Tail | Essential for sperm motility |
| Head | Contains nucleus and acrosome |
| Mitochondria | Energy source for swimming |

**(120) Answer :** (1)
**Solution:**
Actinomorphic flower –Chilli
Zygomorphic flower –Pea, Gulmohur, *Cassia*

**(121) Answer :** (4)
**Solution:**
Baculoviruses can act as biocontrol agents but can not serve the role of biofertilizer.

**(122) Answer :** (1)
**Solution:**
*In-situ* conservation – National park
*Ex-situ* conservation – Zoological park, botanical garden, wildlife safari park

**(123) Answer :** (4)
**Solution:**
Ernst Mayr is known as 'The Darwin of the $20^{th}$ century.

**(124) Answer :** (3)
**Solution:**
RNA though it also acts as a genetic material in some viruses, mostly functions as a messenger. It can also functions as adapter, structural, and in some cases as a catalytic molecule. DNA has evolved from RNA with chemical modifications that make it more stable.

**(125) Answer :** (2)
**Solution:**
Only mRNA. tRNA and rRNA interact with each other during polypeptide synthesis (translation).

**(126) Answer :** (2)
**Solution:**

| | |
|---|---|
| Stomach | Gastrin |
| Pituitary gland | Prolactin |
| Ovary | Relaxin |
| Hypothalamus | Vasopressin |

**(127) Answer :** (1)
**Solution:**
All living members of the class Osteichthyes have bony vertebral column in adults. Claspers are absent in bony fishes. Certain bony fishes such as sea horse and magur (*Clarias*) lack scales. Air sac is not found in fishes but found in birds to assist respiration.

**(128) Answer :** (2)
**Solution:**
Cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum.*

**(129) Answer :** (2)
**Solution:**
In sponges, water canal system is responsible for gathering of food as digestion is intracellular in them. They are stationary in adult forms and there is no locomotion in them. Water canal system is also helpful in removal of waste as well as transportation of male gametes for cross-fertilization because sperms in them mature earlier than ova.

**(130) Answer :** (4)
**Solution:**

| | |
|---|---|
| Rhodophyceae | *Gracilaria* |
| Phaeophyceae | *Ectocarpus* |
| Lycopsida | *Selaginella* |
| Sphenopsida | *Equisetum* |

**(131) Answer :** (2)
**Solution:**

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MRI uses strong magnetic fields and non-ionising radiations. Cellular oncogenes (*c-onc*) when activated under certain conditions, could lead to oncogenic transformation of the cells. Majority of therapeutic drugs have side effects like hair fall, anemia, *etc*.

**(132) Answer :** (2)
**Solution:**
ICSI involves *in-vitro* fertilisation. Parturition is a common method for both babies produced either by test tube baby programme or by naturalconception. Embryo transfer (ET) is employed only in test tube baby programmes.

**(133) Answer :** (2)
**Solution:**
Differentiation of female gametes occurs before the completion of meiosis. Decrease in levels of progesterone during luteal phase of menstrual cycle leads to the start of the first phase of menstrual cycle.

**(134) Answer :** (4)
**Solution:**
Frond is a leaf-like photosynthetic organ, in algae.

**(135) Answer :** (3)
**Solution:**
Materials to be packaged in the form of vesicles from the ER fuse with the *cis* face of the Golgi apparatus and move towards the maturing face. This explains, why the Golgi apparatus remains in close association with the endoplasmic reticulum.

**(136) Answer :** (3)
**Solution:**
According to Katherine Esau's early publications, the curly top virus spreads through a plant via the phloem tissue.

**(137) Answer :** (4)
**Solution:**
After origin, immature lymphocytes mature into antigen-sensitive lymphocytes in primary lymphoid organs such as thymus and bone marrow. After maturation, lymphocytes migrate to secondary lymphoid organs such as spleen, lymph nodes, tonsils, appendix, *etc.*, which provide site for interaction of lymphocytes with antigen.

**(138) Answer :** (4)
**Solution:**

| | |
|---|---|
| Overexploitation | Cause of extinction of Steller's sea cow from Russia |
| On-site conservation | Biodiversity reserves |
| Nile perch | Extinction of cichlid fishes from Lake Victoria |
| Thylacine | Included in the Red list of IUCN as extinct species |

**(139) Answer :** (1)
**Solution:**
In over 60% of angiosperms, pollen grains (mature male gametophyte) are shed at 2-celled stage. In order to form it, one meiosis and one mitosis is required.

**(140) Answer :** (3)
**Solution:**
Innate immunity is a non-specific type of immunity present from birth. Both innate immunity and passive immunity lack memory of the first encounter. Humoral immunity is acquired after birth; it is pathogen-specific and is characterised by the memory of the first encounter.

**(141) Answer :** (4)
**Solution:**
Not all orchids offer rewards to their pollinators. *Ophrys* employs 'sexual deceit' to get pollination done by a species of bee as one of its petal bears an uncanny resemblance to the female bee and male bee pseudocopulates with the flower.

**(142) Answer :** (4)
**Solution:**
Cells of endothecium of anther wall have α-cellulosic fibrous bands arising from inner tangential wall which help in dehiscence of anther due to their hygroscopic nature.

**(143) Answer :** (3)
**Solution:**
In prokaryotes, ribosomes are associated with the plasma membrane of the cell. They are about 15 nm by 20 nm in size and are made of two subunits - 50S and 30S units which when present together form 70S prokaryotic ribosome.

**(144) Answer :** (3)
**Solution:**

Haem is a prosthetic group for enzymes named catalase and peroxidase and is a part of their active sites. Niacin is a vitamin which is found with certain co-enzymes such as NAD and NADP.

**(145) Answer :** (4)
**Solution:**
Hepatic vein carries deoxygenated blood from liver to inferior vena cava. Pulmonary artery carries deoxygenated blood from heart to lungs. Renal portal vein carries deoxygenated blood from lower parts of the body to kidney.

**(146) Answer :** (3)
**Solution:**
In a bryophyte life cycle, the correct sequence of stages will be given as follows:
B → Protonema stage developed from spore
A → Development of lateral bud in secondary protonema
D → Formation of sex organs
E → Transfer of antherozoids to the archegonia in presence of water
C → Development of sporophyte consisting of a seta, foot and capsule

**(147) Answer :** (2)
**Solution:**
Because gene of interest is inserted in the coding sequence of reporter enzyme named β-galactosidase which produces blue colour in the presence of chromogenic substrate, the colour producing ability is lost. So, recombinants will produce white coloured colonies due to insertional inactivation of coding sequence of reporter enzyme whereas non-recombinants will produce blue-coloured colonies.

**(148) Answer :** (4)
**Solution:**
*Trichoderma* is used as biocontrol agent and can also be used for the production of cyclosporin A. *Monascus purpureus* is used for the commercial production of statins. *Nostoc* serves the role of biofertiliser. *Clostridium butylicum* is used for the commercial production of butyric acid. *Acetobacter aceti* is used for the commercial production of acetic acid.

**(149) Answer :** (4)
**Solution:**
In RNAi, nematode specific genes were introduced into the host plant by using *Agrobacterium* vectors. The introduction of DNA was such that it produced both sense and anti-sense RNA in host cells. Thus, two RNAs being complementary to each other formed a dsRNA that initiated RNAi and thus, silenced the specific mRNA of the nematode.

**(150) Answer :** (4)
**Solution:**
NADH dehydrogenase is complex I of mitochondrial ETS.

**(151) Answer :** (2)
**Solution:**
Platyhelminths are triploblastic acoelomates as their body is solid because mesodermal parenchyma is present between body wall and gut wall. They have no body cavity. So, they are considered as triploblastic acoelomates. Because the animal is aquatic, so most probably it is *Planaria.*

**(152) Answer :** (2)
**Solution:**

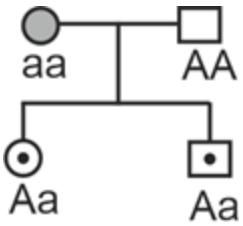
In a floral formula, $\overline{C\ A}$ stands for epipetalous condition of stamens and $\overline{P\ A}$ stands for epiphyllous or epitepalous condition of stamens.

**(153) Answer :** (1)
**Solution:**
Except for the deep sea hydro-thermal ecosystem, sun is the only source of energy for all ecosystems on Earth.
All organisms are dependent for their food on producers, either directly or indirectly.

**(154) Answer :** (2)
**Solution:**

Primary production is expressed as kcal $m^{-2}$.

**(155) Answer :** (3)
**Solution:**
The pedigree shows the inheritance of autosomal recessive trait. Thus, for genotype of $F_3$ generation, the representation can be as follows:




**(156) Answer :** (2)

**Solution:**

In the seeds of cereals, the plumule and radicle are enclosed in sheaths which are called coleoptile and coleorhiza, respectively.

**(157) Answer :** (4)

**Solution:**

OAA - First stable product formed in Hatch and Slack pathway
3-PGA - First stable product formed in Calvin pathway
RuBP - Primary $CO_2$ acceptor in $C_3$ plant
PEP - Primary $CO_2$ acceptor in $C_4$ plant

**(158) Answer :** (3)

**Solution:**

The chemical method developed by Har Gobind Khorana was instrumental in synthesizing RNA molecules with defined combinations of bases.

**(159) Answer :** (3)

**Solution:**

The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called nucleosome.

**(160) Answer :** (4)

**Solution:**

All enzymes such as restriction enzymes, ligases, DNA polymerase and alkaline phosphatase are key tools of recombinant DNA technology. Phosphorylase is used to convert glycogen into glucose- 1-phosphate.

**(161) Answer :** (3)

**Solution:**

In some prokaryotes like cyanobacteria, there are other membranous extensions into the cytoplasm called chromatophores which contain pigments.

**(162) Answer :** (1)

**Solution:**

σ (sigma) factor facilitates the initiation of prokaryotic transcription.

**(163) Answer :** (4)

**Solution:**

Vasa efferentia are 10 to 12 in number that arise from testes. They enter the kidneys on their side and open into Bidder's canal. Finally, Bidder's canal communicates with the urinogenital duct that comes out of the kidneys and opens into the cloaca.

**(164) Answer :** (1)

**Solution:**

Descending limb of loop of Henle is permeable to water but almost impermeable to electrolytes so only small amount of electrolytes such as $Na^+$ is absorbed at this site. Thin ascending limb of loop of Henle is permeable to electrolytes and impermeable to water. So, $Na^+$ and $Cl^-$ are reabsorbed by simple diffusion and filtrate again becomes hypotonic to blood plasma.

**(165) Answer :** (1)

**Solution:**

Antigen binding site is located at N-terminal region of antibody molecule. Each antibody molecule has two portions: one end that binds to antigens is called Fab portion which is situated towards sub-script N-terminal and other portion which is responsible for its effector function like neutralisation, opsonisation, *etc.,* is called FC portion which is situated towards C-terminals of polypeptide chains. Each light and heavy chain is connected by single disulphide bond whereas two heavy chains are held together by two disulphide bonds.

**(166) Answer :** (3)

**Solution:**

| | |
|---|---|
| Epicotyl | It has shoot apex and a few leaf primordia |
| Hypocotyl | Terminates at its lower end in the radicle |
| Endosperm | Not found in the mature seeds of orchids |
| Hilum | Scar on the seed coat |

**(167) Answer :** (3)

**Solution:**

Pericycle is present on the inner side of the endodermis and above the phloem in the form of semi-lunar patches of sclerenchyma.

**(168) Answer :** (3)

**Solution:**

Both are identical twins derived from a single zygote which cleaves to form a single blastula that gets seperated after implantation. So, they have 100% identical genetic contents. They are called maternal/identical monozygotic twins. In IVF, embryo with more than 8 blastomeres are implanted in uterus.

**(169) Answer :** (4)

**Solution:**

Forelimbs of mammals such as human, whale, bat, horse, *etc.,* show homology. Homologous organs evolve due to divergent evolution and analogous organs evolve due to convergent evolution.

**(170) Answer :** (2)

**Solution:**

Auxin controls xylem differentiation and helps in cell division.

**(171) Answer :** (2)

**Solution:**

Insulin injection is usually taken before 0-30 minutes of taking meal as it synchronizes the insulin's peak action with the rise in blood glucose levels after taking food resulting in better glycemic control. Insulin is a quaternary protein containing A and B chains. It is digested by digestive enzymes, and thus, it cannot be taken orally.

**(172) Answer :** (3)

**Solution:**

Lyases are class of enzymes that catalyse removal of groups from substrates by mechanism other than hydrolysis leaving double bonds. So, option (3) represents enzymatic reaction catalysed by a lyase. Reaction (1) is catalysed by oxido-reductase, reaction (2) is catalysed by a transferase and reaction (4) is catalysed by a DNA ligase.

**(173) Answer :** (1)

**Solution:**

The $\mathrm{DNA}$ fragments separate (resolve) according to their size through sieving effect provided by agarose gel.

**(174) Answer :** (1)

**Solution:**

The correct sequence of events in the life cycle of pteridophytes is B → E → C → A → D

**(175) Answer :** (1)

**Solution:**

In the classical Mendelian dihybrid cross the phenotypic ratio in $F_2$ generation comes out to be 9 : 3 : 3 : 1 and genotypic ratio comes out to be 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1

**(176) Answer :** (4)

**Solution:**

Increasing order of complexity of body organisation in the kingdom given by R.H. Whittaker is as follows:
C → E→ A → D → B

**(177) Answer :** (2)

**Solution:**

| | |
|---|---|
| Centriole | Cartwheel organisation |
| Capsule | Thick and tough glycocalyx |
| Cisternae | Component of golgi apparatus |
| Chromoplast | Contain fat soluble pigments |

**(178) Answer :** (1)

**Solution:**

Exponential growth can be expressed as $\frac{dN}{dt} = rN$.

**(179) Answer :** (3)

**Solution:**

| | |
|---|---|
| Uremia | Accumulation of urea in blood |
| Muscular dystrophy | Progressive degeneration of skeletal muscles |
| Heart attack | Heart muscle is suddenly damaged due to inadequate blood supply |
| Asthma | Difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles |

**(180) Answer :** (4)

**Solution:**

Increase in $pCO_2$, decrease in pH, increase in $H^+$ concentration in blood/body and activation of pneumotaxic centre present in pons are events in our body which activate respiratory rhythm centre to increase breathing rate. Decreased alevolar $CO_2$ and metabolic alkalosis decrease breathing rate by inhibiting respiratory rhythm centre.

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